

**Theorem** (Jensen's inequality). *Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be convex, i.e.*

$$\forall x, y \in \mathbb{R}, \forall t \in [0, 1] \quad f(tx + (1-t)y) \leq tf(x) + (1-t)f(y). \quad (\text{C})$$

*Let  $x_1, \dots, x_n \in \mathbb{R}$  and weights  $\lambda_1, \dots, \lambda_n$  satisfy  $\lambda_i \geq 0$  and  $\sum_{i=1}^n \lambda_i = 1$ . Then*

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i). \quad (\text{J})$$

*Proof.* **1. Weight conditions.** The hypotheses imply that

$$X := \sum_{i=1}^n \lambda_i x_i$$

is a convex combination of the numbers  $x_i$ .

- 2. Base case  $n = 2$ .** For two points write  $\lambda_1 = t$ ,  $\lambda_2 = 1 - t$  with  $t \in [0, 1]$ . By (C),

$$f(tx_1 + (1-t)x_2) \leq tf(x_1) + (1-t)f(x_2),$$

which is exactly (J) for  $n = 2$ .

- 3. Induction hypothesis.** Assume that for a fixed  $k \geq 2$  the statement (J) holds for every collection of  $k$  points and any non-negative weights summing to one; i.e. for any  $y_1, \dots, y_k$  and any  $\mu_1, \dots, \mu_k$  with  $\mu_i \geq 0$  and  $\sum_{i=1}^k \mu_i = 1$ ,

$$f\left(\sum_{i=1}^k \mu_i y_i\right) \leq \sum_{i=1}^k \mu_i f(y_i). \quad (\text{IH})$$

- 4. Prepare the  $(k+1)$ -point case.** Let  $x_1, \dots, x_{k+1}$  and weights  $\lambda_1, \dots, \lambda_{k+1}$  satisfy the same conditions. Set

$$\alpha := \lambda_{k+1}, \quad \beta := 1 - \alpha = \sum_{i=1}^k \lambda_i.$$

Because the  $\lambda_i$  are non-negative,  $\beta \in [0, 1]$ . If  $\beta = 0$  then  $\lambda_{k+1} = 1$  and the desired inequality reduces to the trivial equality  $f(x_{k+1}) = f(x_{k+1})$ . Otherwise define

$$y := \frac{1}{\beta} \sum_{i=1}^k \lambda_i x_i. \quad (4)$$

The coefficients  $\lambda_i/\beta$  are non-negative and sum to one, so (4) is a convex combination of the first  $k$  points.

- 5. Apply the induction hypothesis to the first  $k$  points.** Using (IH) with the points  $x_1, \dots, x_k$  and the weights  $\lambda_i/\beta$  we obtain

$$f(y) = f\left(\sum_{i=1}^k \frac{\lambda_i}{\beta} x_i\right) \leq \sum_{i=1}^k \frac{\lambda_i}{\beta} f(x_i). \quad (5)$$

- 6. Rewrite the whole convex combination as a two-point convex combination.** Directly from the definitions,

$$\begin{aligned}
X &:= \sum_{i=1}^{k+1} \lambda_i x_i \\
&= \beta \left( \frac{1}{\beta} \sum_{i=1}^k \lambda_i x_i \right) + \alpha x_{k+1} \\
&= \beta y + \alpha x_{k+1}.
\end{aligned} \tag{6}$$

Here  $\beta, \alpha \in [0, 1]$  and  $\beta + \alpha = 1$ .

- 7. Apply convexity to the pair  $(y, x_{k+1})$ .** Using (C) with  $t = \beta$  gives

$$f(X) = f(\beta y + \alpha x_{k+1}) \leq \beta f(y) + \alpha f(x_{k+1}). \tag{7}$$

- 8. Insert the bound (5) for  $f(y)$  into (7).** Combining (5) and (7),

$$\begin{aligned}
f(X) &\leq \beta \left( \sum_{i=1}^k \frac{\lambda_i}{\beta} f(x_i) \right) + \alpha f(x_{k+1}) \\
&= \sum_{i=1}^k \lambda_i f(x_i) + \lambda_{k+1} f(x_{k+1}) \\
&= \sum_{i=1}^{k+1} \lambda_i f(x_i).
\end{aligned} \tag{8}$$

This is precisely the desired inequality for  $k + 1$  points.

- 9. Induction conclusion.** The base case  $n = 2$  holds, and the induction step shows that truth for  $k$  implies truth for  $k+1$ . Hence, by mathematical induction, (J) is valid for every integer  $n \geq 2$ . The case  $n = 1$  is immediate because both sides equal  $f(x_1)$ .
- 10. Equality condition.** Equality in (C) occurs exactly when either the weight is 0 or 1 (so one point is not used) or when  $f$  is affine on the segment joining the two points. Propagating this through the induction argument yields equality in Jensen's inequality precisely when all points with positive weight are equal, or when  $f$  is affine on the convex hull of  $\{x_1, \dots, x_n\}$ .

Thus, for any convex function  $f$  and any convex combination of real numbers, Jensen's inequality holds.  $\square$